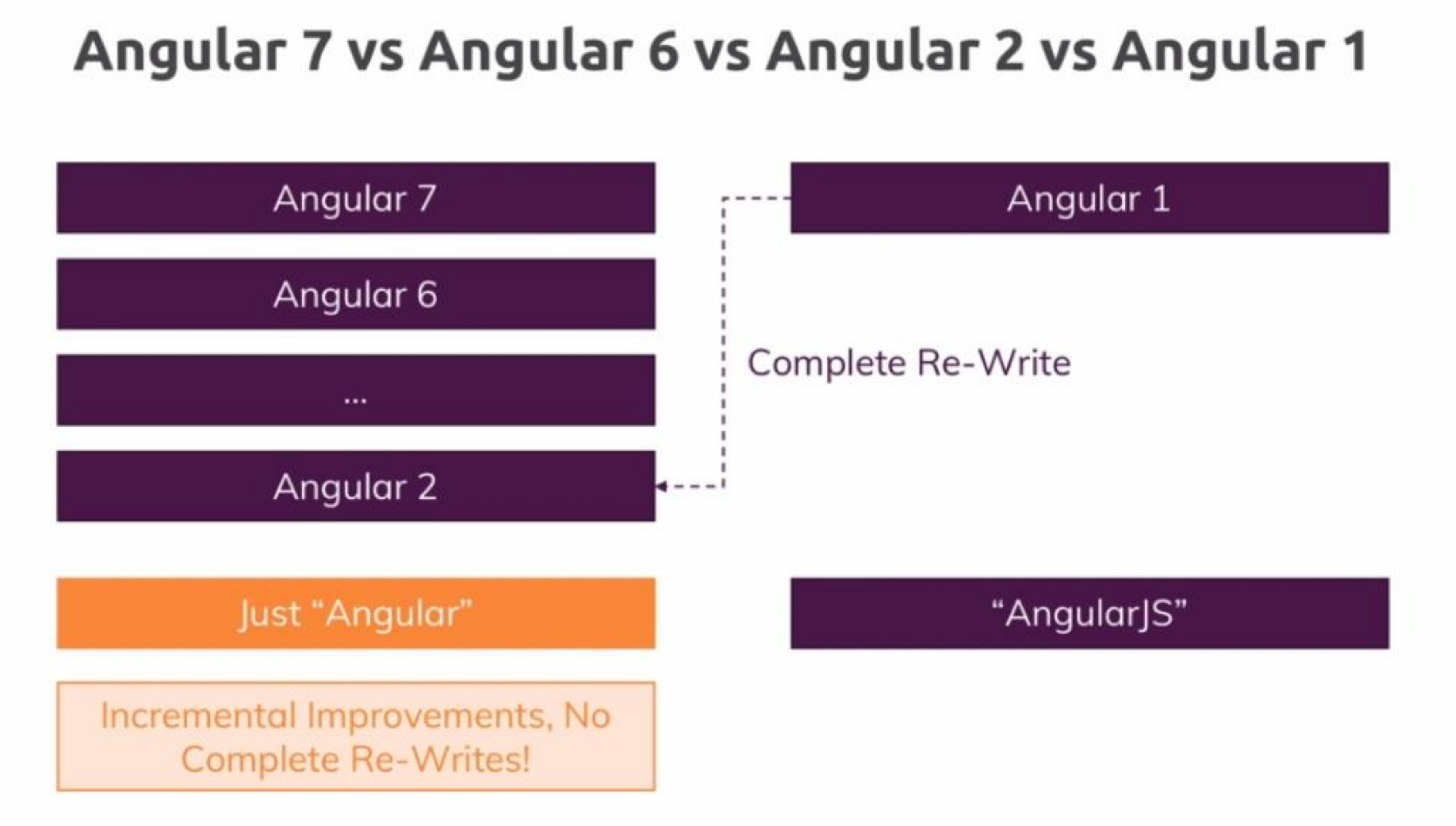
ANGULAR

By: Michel Samir Zaki

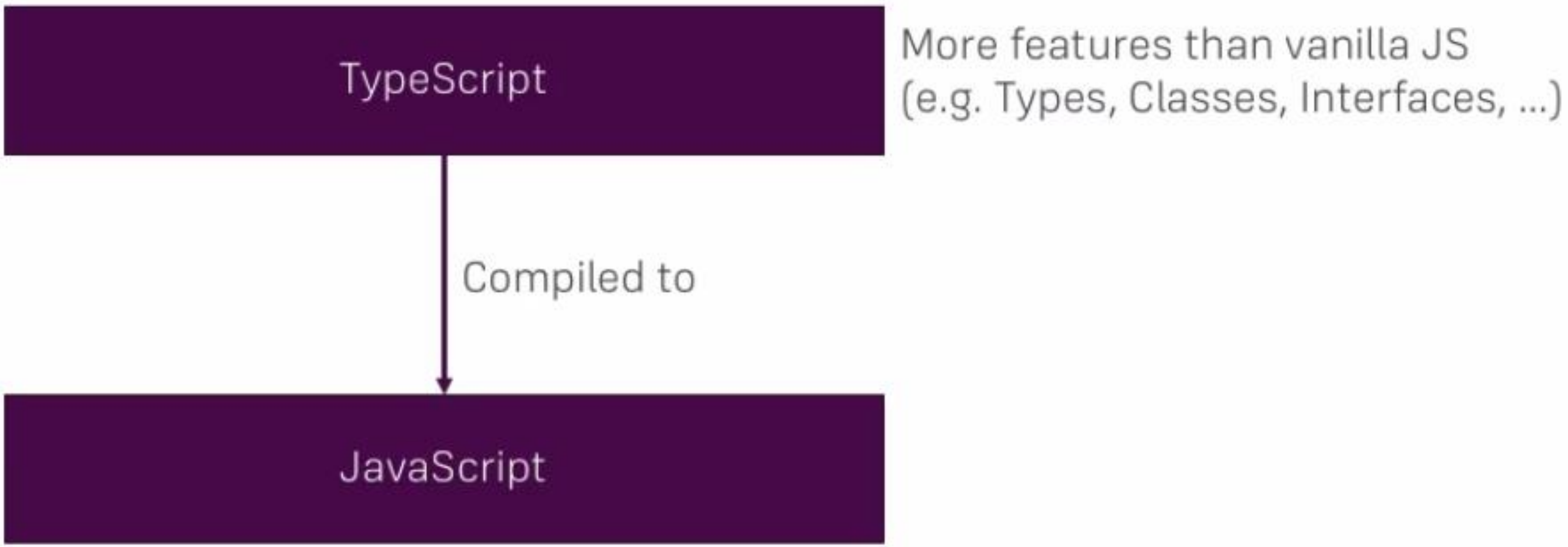
What is Angular

Angular is a **JavaScript Framework** which allows you to create reactive **Single Page Application**

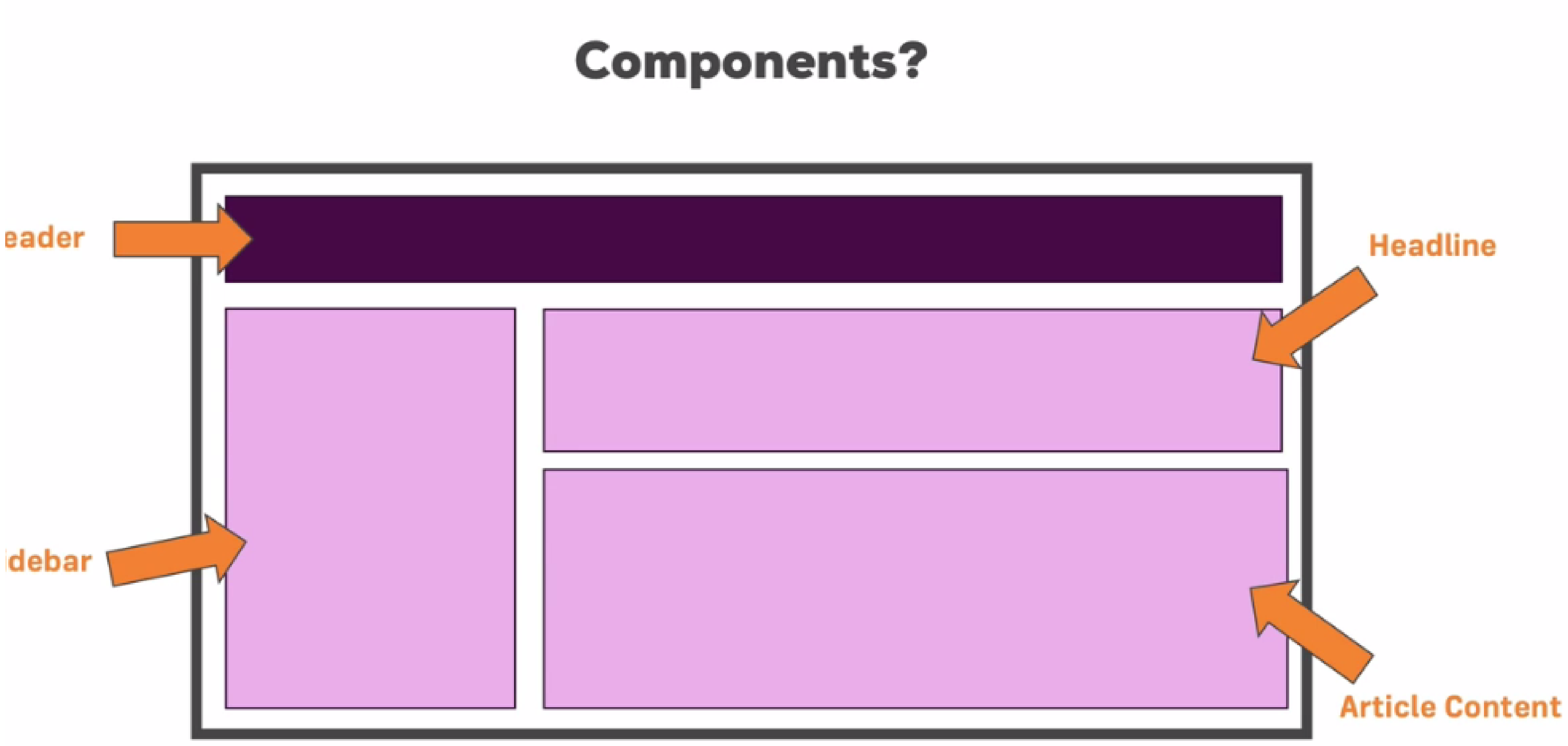
What is Angular



TypeScript

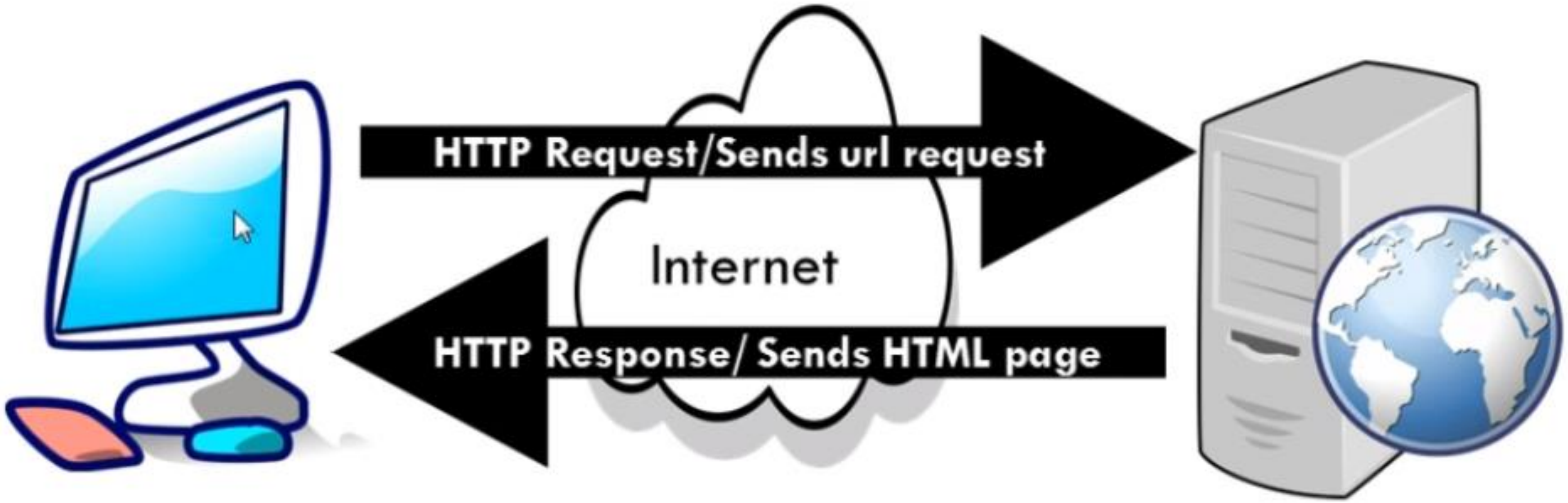

TypeScript
More features than vanilla JS
(e.g. Types, Classes, Interfaces, ...)
Compiled to
JavaScript

Components



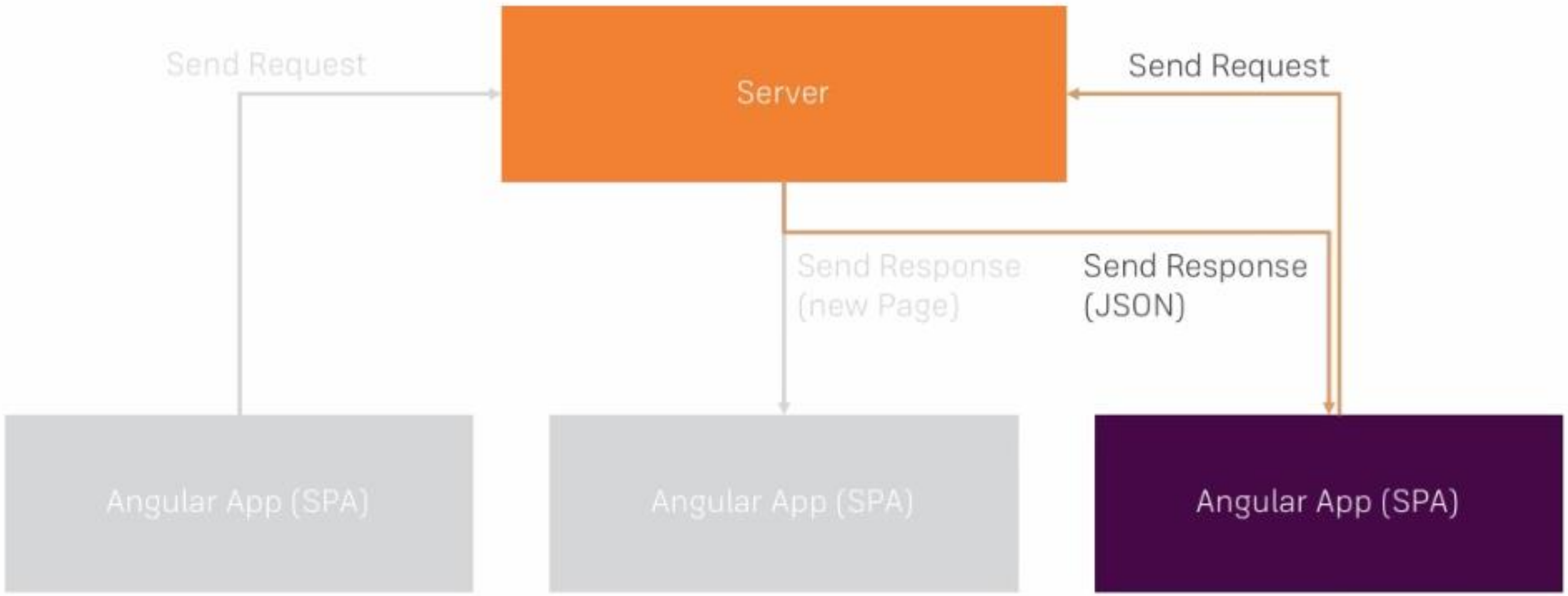
SPA VS MPA client/server application

- MPA (Client/Server App)



SPA VS MPA client/server application

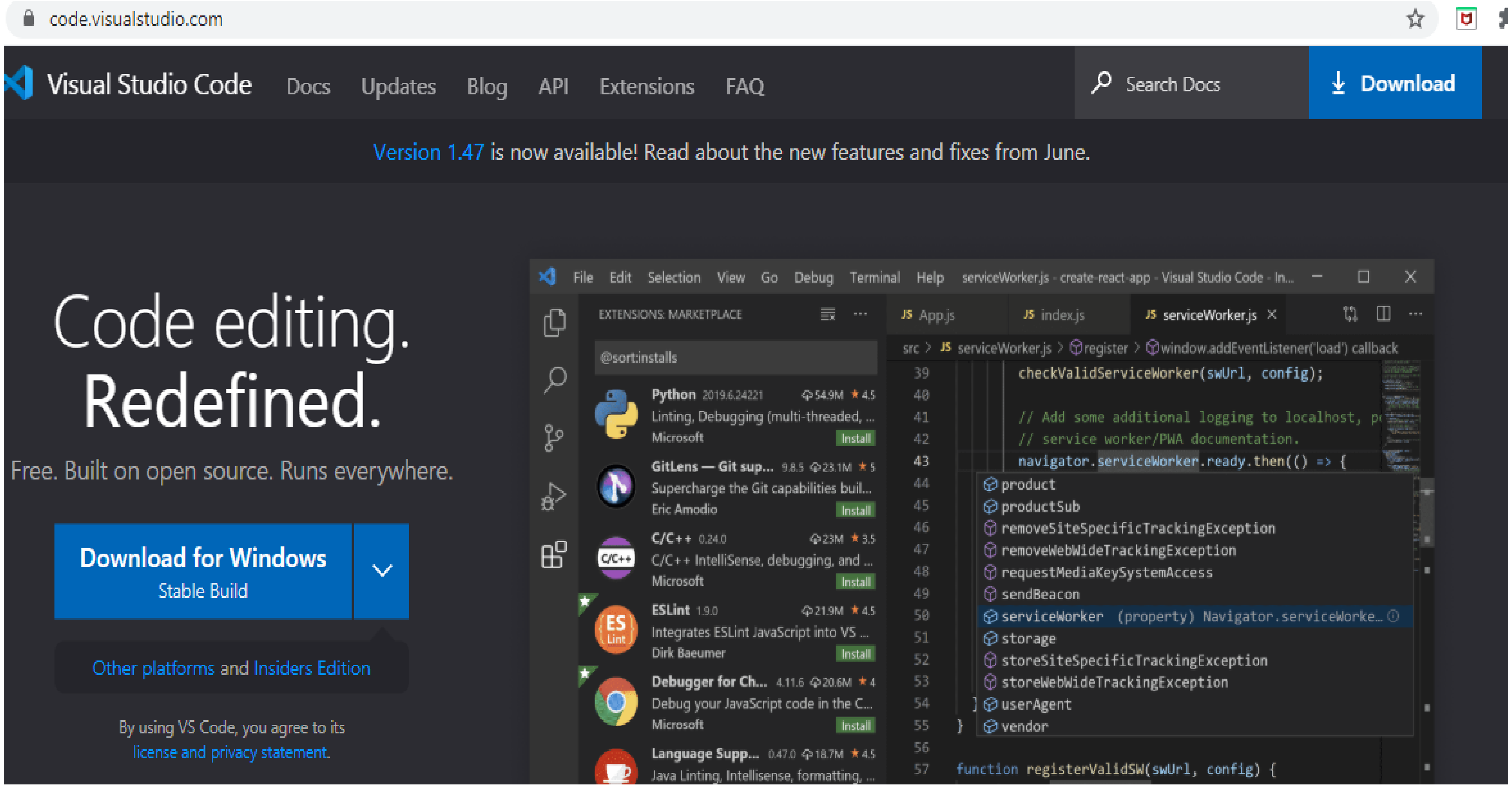
- SPA (Single Page App)



Programs Needed for Course

- **Visual Studio Code (Editor which I will use)**

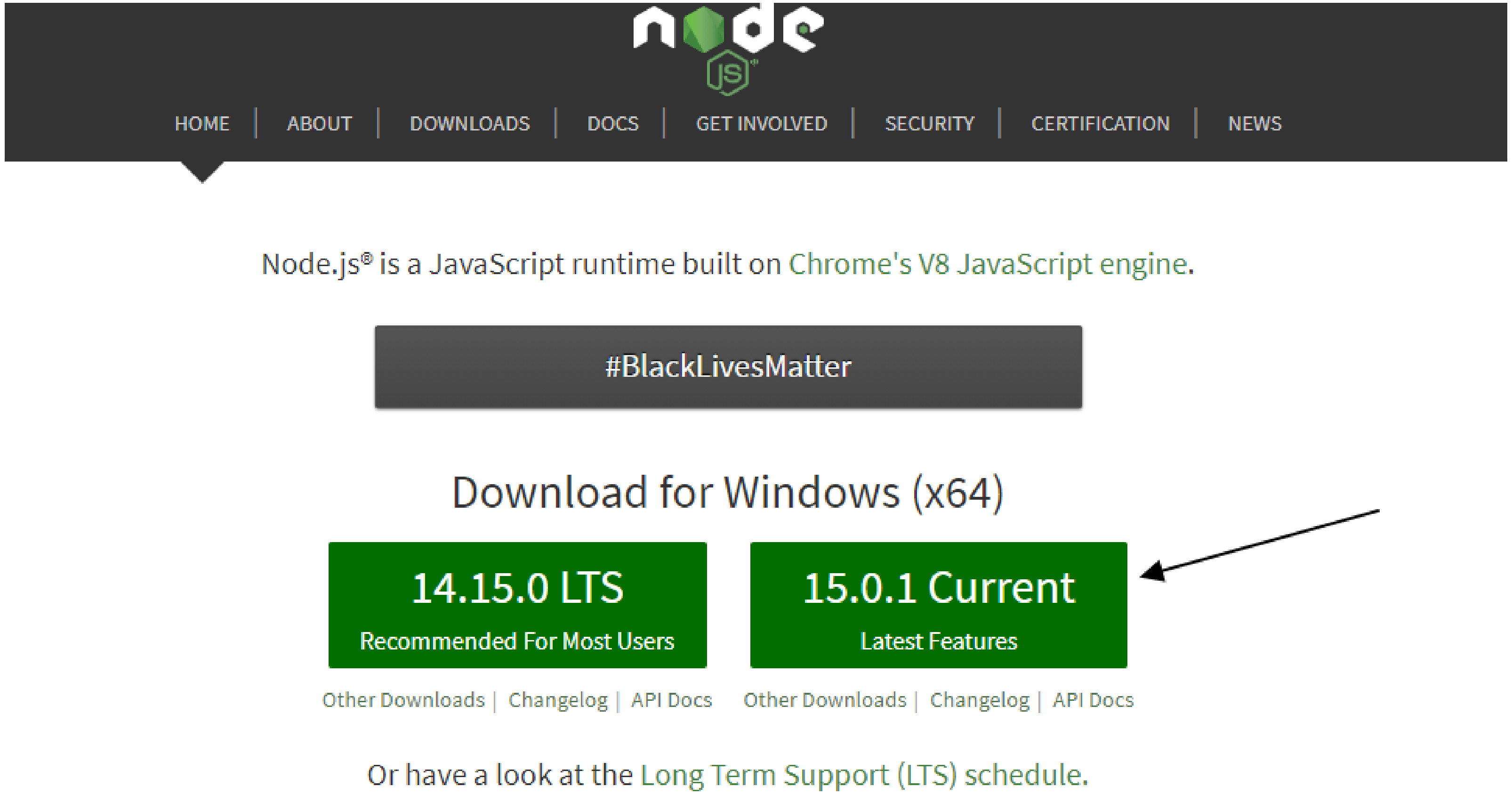
https://code.visualstudio.com/



Programs Needed for Course

- **Node Js (**JavaScript runtime – server side**)**

https://nodejs.org/en/



Programs Needed for Course

- **Angular (CLI)**

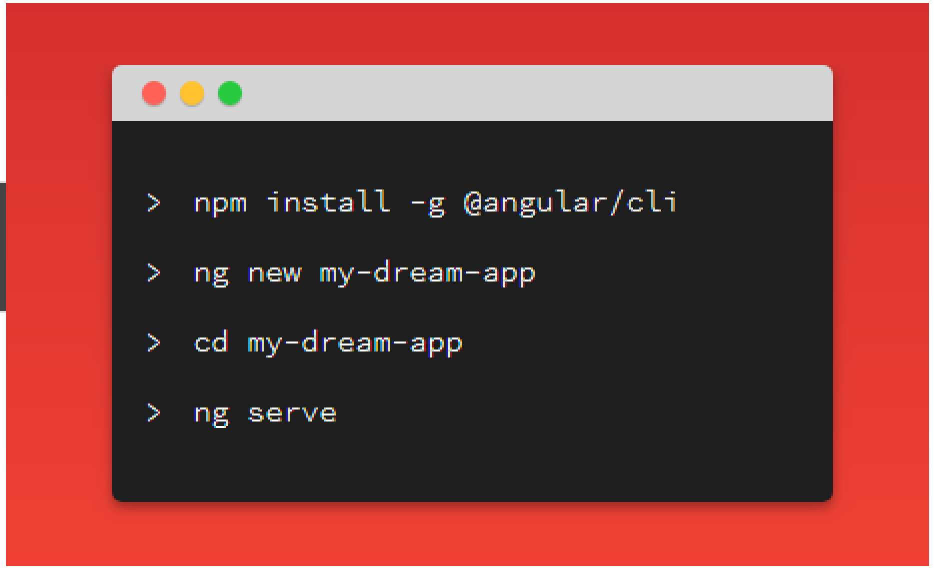
https://angular.io/cli

Install the Angular CLI

You use the Angular CLI to create projects, generate application and library code, and perform a variety of ongoing development tasks such as testing, bundling, and deployment.

To install the Angular CLI, open a terminal window and run the following command:

```
npm install -g @angular/cli
```



Create New Project

Create New Project

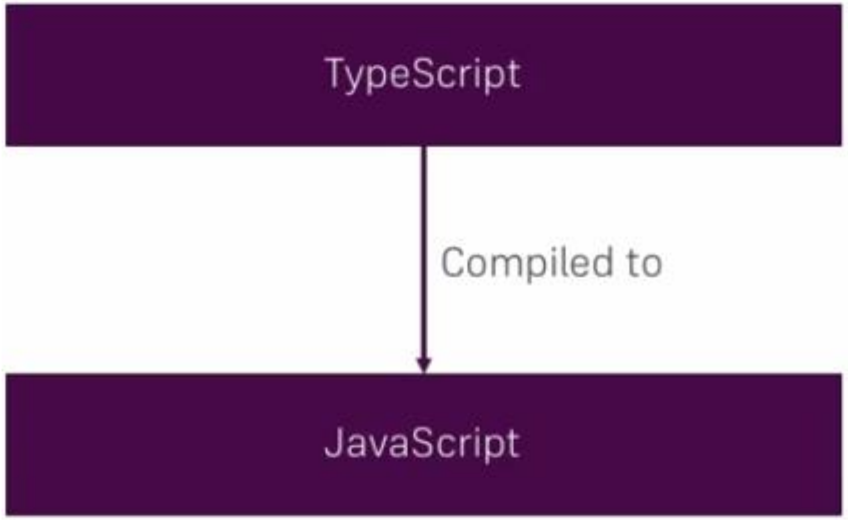
- Yes or now will not matter now

```
npm
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

D:\faculty\faculty\SW2\Dr Amr Goniem\Weeks\Week 1\Code>ng new angular-app
? Would you like to add Angular routing? No
? Which stylesheet format would you like to use? CSS
```

How Angular app runs

```
PROBLEMS   TERMINAL   OUTPUT   DEBUG CONSOLE

Session contents restored from 3/2/2022 at 3:17:36 PM

Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

D:\faculty\faculty\SW2\Dr Amr Goniem\Weeks\Week 1\Code\section1-week1>
ng serve
```

```
PROBLEMS   TERMINAL   OUTPUT   DEBUG CONSOLE

Date: 2022-03-03T07:06:02.981Z
Hash: e7c8d570092d14bba406
Time: 14562ms
chunk {main} main.js, main.js.map (main) 9.93 kB [initial] [rendered]
chunk {polyfills} polyfills.js, polyfills.js.map (polyfills) 237 kB [initial] [
chunk {runtime} runtime.js, runtime.js.map (runtime) 6.08 kB [entry] [rendered]
chunk {styles} styles.js, styles.js.map (styles) 16.3 kB [initial] [rendered]
chunk {vendor} vendor.js, vendor.js.map (vendor) 3.44 MB [initial] [rendered]
i ｢wdm｣: Compiled successfully.
```



Index.html

```html
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Section1Week1</title>
  <base href="/">

  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="icon" type="image/x-icon" href="favicon.ico">
</head>
<body>
  <app-root></app-root>
<script type="text/javascript" src="runtime.js"></script><script type="text/javascript" src=
</html>
```

How Angular app runs

main.ts

```typescript
import { enableProdMode } from '@angular/core';
import { platformBrowserDynamic } from '@angular/platform-

import { AppModule } from './app/app.module';
import { environment } from './environments/environment';

if (environment.production) {
  enableProdMode();
}

platformBrowserDynamic().bootstrapModule(AppModule)
  .catch(err => console.error(err));
```

AppModule

```typescript

import { AppComponent } from './app.compor

@NgModule({
  declarations: [
    AppComponent,
  ],
  imports: [
    BrowserModule
  ],
  providers: [],
  bootstrap: [AppComponent]
})
export class AppModule { }
```

AppComponent

```typescript
import { Component } from '@angular/core';

@Component({
  selector: 'app-root',
  templateUrl: './app.component.html',
  styleUrls: ['./app.component.css']
})
export class AppComponent {
  title = 'section1-week1';
}
```

App Component in angular (root component)

index.html

```
index.html ×
src > index.html > ...
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Section1Week1</title>
  <base href="/">

  <meta name="viewport" content="wi
  <link rel="icon" type="image/x-ic
</head>
<body>
  <app-root></app-root>
</body>
</html>
```

AppComponent

```
SECTION1-WEEK1
> e2e
> node_modules
v src
  v app
      app.component.css
      app.component.html          M
      app.component.spec.ts
      app.component.ts
      app.module.ts               M
  > assets
  > environments
    browserslist
    favicon.ico
```

AppComponent

```
app.component.ts ×
src > app > app.component.ts > ...
import { Component } from '@angular/core';

@Component({
  selector: 'app-root',
  templateUrl: './app.component.html',
  styleUrls: ['./app.component.css']
})
export class AppComponent {
  title = 'section1-week1';
}
```


Thank You